

Electrodynamics & Optics

Part II Physics — Course Summary

Compiled from Handouts 1–3, the lecture overheads and the examples sheet
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This document summarises the key concepts, definitions and results of the course. Full derivations of the central results are given in the companion document “Key Derivations.” Sections are organised to follow the lecture handouts. SI units are used throughout; a time dependence $e^{-i\omega t}$ is adopted.

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1 Electromagnetic Waves

1.1 Maxwell's equations and media

The Lorentz force on a charge q is $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. In a medium the response is described by the polarization $\mathbf{P}$ (bound charge $\rho_p = -\nabla \cdot \mathbf{P}$, bound current $\mathbf{J}_p = \dot{\mathbf{P}}$) and magnetization $\mathbf{M}$ (bound current $\mathbf{J}_m = \nabla \times \mathbf{M}$). Introducing the auxiliary fields

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad \mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M},$$

Maxwell's equations take their standard macroscopic form (free sources only):

$\nabla \cdot \mathbf{D} = \rho$	(ME1)	$\nabla \cdot \mathbf{B} = 0$	(ME2)
$\nabla \times \mathbf{E} = -\dot{\mathbf{B}}$	(ME3)	$\nabla \times \mathbf{H} = \mathbf{J} + \dot{\mathbf{D}}$	(ME4)

For linear, isotropic media the constitutive relations are $\mathbf{D} = \epsilon \epsilon_0 \mathbf{E}$ and $\mathbf{B} = \mu \mu_0 \mathbf{H}$, with relative permittivity $\epsilon = 1 + \chi$ and permeability $\mu = 1 + \chi_m$.

1.2 Boundary conditions

From the integral forms: $D_\perp$ continuous (no free surface charge), $B_\perp$ continuous, $E_\parallel$ continuous, $H_\parallel$ continuous (no free surface current).

1.3 EM energy and the Poynting vector

Energy density and flux:

$$u = \frac{1}{2} \mathbf{E} \cdot \mathbf{D} + \frac{1}{2} \mathbf{B} \cdot \mathbf{H}, \quad \mathbf{N} = \mathbf{E} \times \mathbf{H}.$$

Poynting's theorem $\frac{dU}{dt} = -\oint_S \mathbf{N} \cdot d\mathbf{S} - \int_V \mathbf{J} \cdot \mathbf{E} dV$ expresses conservation of energy as a radiated flux plus an ohmic dissipation term.

1.4 The vector potential

Since $\nabla \cdot \mathbf{B} = 0$, $\mathbf{B} = \nabla \times \mathbf{A}$; ME3 then gives $\mathbf{E} = -\dot{\mathbf{A}} - \nabla \phi$. In time-varying situations $\mathbf{E}$ is *not* conservative and both potentials are needed.

1.5 Plane EM waves

In a source-free region each field obeys the wave equation $\nabla^2 \mathbf{E} = \epsilon \epsilon_0 \mu \mu_0 \ddot{\mathbf{E}}$, giving phase speed

$$v = \frac{1}{\sqrt{\epsilon \epsilon_0 \mu \mu_0}} = \frac{c}{\sqrt{\epsilon \mu}} = \frac{c}{n}, \quad c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}.$$

For harmonic plane waves $e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$: $\mathbf{k} \cdot \mathbf{D} = 0$, $\mathbf{k} \cdot \mathbf{B} = 0$, $\mathbf{k} \times \mathbf{E} = \omega \mathbf{B}$, $\mathbf{k} \times \mathbf{H} = -\omega \mathbf{D}$. Hence $\mathbf{B}, \mathbf{D} \perp \mathbf{k}$; for isotropic media $(\mathbf{E}, \mathbf{H}, \mathbf{k})$ are mutually perpendicular and $\mathbf{N} \parallel \mathbf{k}$. Wave impedance:

$$\frac{|\mathbf{E}|}{|\mathbf{H}|} = \sqrt{\frac{\mu \mu_0}{\epsilon \epsilon_0}} = \sqrt{\frac{\mu}{\epsilon}} Z_0, \quad Z_0 = \sqrt{\mu_0 / \epsilon_0} \approx 377 \Omega.$$

A complex wavevector $\mathbf{k} \rightarrow \mathbf{k} + i\boldsymbol{\kappa}$ describes damped (evanescent) waves; if $\boldsymbol{\kappa} \parallel \mathbf{k}$ the wave is homogeneous. Longitudinal waves are possible where $\epsilon = 0$ (e.g. plasma oscillations at ω_p).

1.6 Fresnel coefficients and the Brewster angle

At a planar interface the reflection/transmission amplitudes split into p (E in plane of incidence) and s (E perpendicular) cases:

$$r_p = \frac{n_2 \cos \theta_1 - n_1 \cos \theta_2}{n_2 \cos \theta_1 + n_1 \cos \theta_2}, \quad r_s = \frac{n_1 \cos \theta_1 - n_2 \cos \theta_2}{n_1 \cos \theta_1 + n_2 \cos \theta_2},$$

with corresponding t_p, t_s . Reflection of the p wave vanishes at the Brewster angle $\theta_B = \tan^{-1}(n_2/n_1)$.

2 Optics: Polarization, Anisotropic Media and Coherence

2.1 Polarization states and Jones calculus

Superposing two perpendicular linearly-polarized waves $\mathbf{E}_T = (a_1 \hat{\mathbf{i}} + a_2 \hat{\mathbf{j}}) E_0 e^{i(kz - \omega t)}$ gives: linear (real a_1, a_2), circular ($a_2 = \pm i a_1$) or elliptical (general). A polarization state is a Jones vector; optical elements are 2×2 Jones matrices, combined as $J = J_C J_B J_A$ (last element applied first):

$$\begin{array}{ll} x\text{-polarizer} & \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \theta\text{-polarizer} \begin{pmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{pmatrix} \\ \lambda/4 \text{ plate } (x \text{ fast}) & \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad \lambda/2 \text{ plate } (x \text{ fast}) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{array}$$

Malus's law $I(\theta) = I_0 \cos^2 \theta$ follows immediately. Crossed polarizers give $J = 0$ (no transmission).

2.2 Anisotropic media (birefringence)

In an anisotropic crystal the permittivity is a tensor, $D_i = \epsilon_{ij} E_j$, so in general $\mathbf{D} \nparallel \mathbf{E}$. Energy-flow arguments show ϵ_{ij} is **Hermitian** ($\epsilon_{ij} = \epsilon_{ji}^*$); for lossless, non-active media it is real and symmetric and can be diagonalised onto principal axes with principal indices n_1, n_2, n_3 :

- **biaxial** ($n_1 \neq n_2 \neq n_3$);
- **uniaxial** ($n_1 = n_2 = n_o \neq n_e = n_3$): birefringence $\Delta n = n_e - n_o$ (calcite: $\Delta n = -0.172$).

The *optical indicatrix* $\sum D_i^2/\epsilon_i = 1$ encodes the refractive index for a given $\mathbf{D}$ direction; the radius vector in a direction equals the index for a wave polarized along it. Because $\mathbf{N} = \mathbf{E} \times \mathbf{H}$ need not be parallel to $\mathbf{k}$, phase and energy can propagate in different directions.

2.3 Waveplates (retarders)

A plate of thickness d with fast/slow indices n_f, n_s imposes a relative phase $\Delta\phi = \omega(n_s - n_f)d/c$ between the principal axes; Jones matrix $\propto \text{diag}(e^{-i\Delta\phi/2}, e^{i\Delta\phi/2})$. $\Delta\phi = \pi/2$ gives a $\lambda/4$ plate (linear $\rightarrow$ elliptical/circular), $\Delta\phi = \pi$ a $\lambda/2$ plate (rotates the plane of polarization by -2θ).

2.4 Induced birefringence

Photoelasticity: stress makes an isotropic solid birefringent (stress visualisation between crossed polarizers). **Kerr effect** (liquids/gases): $\Delta n = \lambda_0 K E_0^2$ (quadratic; no inversion-symmetry distinction). **Pockels effect** (non-centrosymmetric crystals): Δn linear in E_0 . Both enable fast voltage-controlled waveplates/modulators.

2.5 Optical activity and the Faraday effect

Chiral media respond differently to LCP and RCP (the characteristic states), with indices n_L, n_R . A plane wave's polarization rotates by $\Delta\phi/2$; the specific rotatory power is

$$\alpha = \frac{\pi(n_L - n_R)}{\lambda} = \frac{\omega}{2c}(n_L - n_R).$$

The **Faraday effect** induces chirality by an applied $\mathbf{B}_0$. For a magnetised plasma (Faraday geometry $\mathbf{k} \parallel \mathbf{B}_0 \parallel \hat{z}$) the electron equation of motion gives $\epsilon_{L,R}(\omega) = 1 - \omega_p^2/[\omega(\omega \mp \omega_c)]$, with cyclotron frequency $\omega_c = eB_0/m$. The rotation $\theta = VB_0d$ defines the Verdet coefficient V , and the dielectric tensor becomes Hermitian with off-diagonal (magnetically-induced chiral) terms.

2.6 Interference and partial polarization

Perpendicularly polarized waves do not interfere. Degree of polarization $V = I_{\text{pol}}/(I_{\text{pol}} + I_{\text{unpol}})$. **Fresnel–Arago laws**: (i) parallel-polarized coherent beams interfere; (ii) perpendicular polarizations never interfere; (iii) beams from perpendicular components of unpolarized light are mutually incoherent.

2.7 Coherence

Real sources are quasi-monochromatic. Two complementary regimes:

Temporal (longitudinal) coherence

Probed by amplitude division (Michelson interferometer, delay $\tau = 2d/c$). Temporal coherence function $\Gamma(\tau) = \langle f(t)f^*(t - \tau) \rangle$; measured intensity $I(\tau) = 2I_0 + 2\Re[\Gamma(\tau)]$; visibility $V(\tau) = |\Gamma(\tau)|/I_0 = |\gamma(\tau)|$. The **Wiener–Khinchine theorem** gives the power spectrum as the FT of the autocorrelation: $P(\omega) \sim \text{FT}[\gamma(\tau)]$ — the basis of **Fourier-transform spectroscopy** (Jacquinot & Fellgett advantages). A Gaussian line of width σ gives $V(\tau) = \exp(-\sigma^2\tau^2/2)$ and coherence length $l_c = c/(\sqrt{2}\sigma) \sim \lambda^2/(2\delta\lambda)$.

Spatial (lateral) coherence

Probed by wavefront division (Young’s slits). The **van Cittert–Zernike theorem**: the degree of lateral coherence is the (normalised) FT of the source’s angular intensity distribution,

$$\gamma(u) = \frac{1}{I_0} \int I(\theta) e^{-iu\theta} d\theta = \frac{\text{FT}[I(\theta)]}{I_0}, \quad u = kd.$$

A uniform line source gives $\gamma = \text{sinc}(u\alpha/2)$ (first zero at $d = \lambda/\alpha$); a uniform disc gives $\gamma = 2J_1(u\alpha/2)/(u\alpha/2)$ and coherence width $w_c \approx 1.22\lambda/\alpha$. The broader the source, the narrower w_c .

Synthetic-aperture imaging

Michelson’s stellar interferometer measures fringe visibility vs. baseline to resolve stellar diameters (e.g. Betelgeuse). Each baseline samples one Fourier component of the source brightness; **phase closure** over three or more apertures recovers the true phase, underpinning modern interferometric imaging and radio astronomy (VLBI, ALMA).

3 Electrodynamics: Potentials and the Aharonov–Bohm Effect

3.1 The vector potential and gauge freedom

$\mathbf{B} = \nabla \times \mathbf{A}$ implies the flux through a loop L equals $\oint_L \mathbf{A} \cdot d\mathbf{L}$. $\mathbf{A}$ is not unique: under a **gauge transformation**

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi, \quad \phi \rightarrow \phi - \dot{\chi},$$

$\mathbf{E}, \mathbf{B}$ are unchanged. Useful choices: **Coulomb gauge** $\nabla \cdot \mathbf{A} = 0$; **Lorenz gauge** $\nabla \cdot \mathbf{A} + (\epsilon\mu/c^2)\dot{\phi} = 0$.

3.2 Calculating $\mathbf{A}$ (standard results)

Methods: direct integration of $\mathbf{J}$ (Biot–Savart), integrating a known $\mathbf{B}$, or equating $\oint \mathbf{A} \cdot d\mathbf{L}$ to the enclosed flux.

- Long straight wire: $A_z = -\frac{\mu_0 I}{2\pi} \ln r + \text{const.}$

- Distant current loop (magnetic dipole $\mathbf{m}$): $\mathbf{A} = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3}$, with $\mathbf{m} = IA\hat{\mathbf{n}}$.
- Infinite solenoid: $A_\phi = \mu_0 n I r / 2$ inside, $A_\phi = \mu_0 n I a^2 / (2r)$ outside — note $\mathbf{A} \neq 0$ outside even though $\mathbf{B} = 0$.

3.3 $\mathbf{A}$ in quantum mechanics; the Aharonov–Bohm effect

The Hamiltonian uses potentials, $H = \frac{1}{2m}[\mathbf{p}_c - q\mathbf{A}]^2 + q\phi$ (kinetic vs. canonical momentum). Gauge invariance of the Schrödinger equation requires the wavefunction to acquire a phase $\Psi' = e^{iq\chi/\hbar}\Psi$. For two electron paths enclosing a flux Φ_B (with $\mathbf{B} = 0$ everywhere on the paths), the relative phase is

$$\Delta = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\mathbf{s} = \frac{q\Phi_B}{\hbar},$$

shifting the interference pattern — the **Aharonov–Bohm effect** (Tonomura’s experiment). In a superconducting environment the flux is quantised in units $\Phi_0 = h/2e$ (Meissner expulsion), giving $\Delta = n\pi$.

3.4 Maxwell’s equations for $\mathbf{A}$, ϕ ; retarded potentials

In the Lorenz gauge the potentials satisfy decoupled wave equations with source terms:

$$\frac{\epsilon\mu}{c^2} \ddot{\phi} - \nabla^2 \phi = \frac{\rho}{\epsilon\epsilon_0}, \quad \frac{\epsilon\mu}{c^2} \ddot{\mathbf{A}} - \nabla^2 \mathbf{A} = \mu\mu_0 \mathbf{J}.$$

Their causal solutions are the **retarded potentials**

$$\phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{[\rho]}{|\mathbf{r} - \mathbf{r}'|} dV', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{[\mathbf{J}]}{|\mathbf{r} - \mathbf{r}'|} dV',$$

where $[\cdot]$ means evaluation at the retarded time $t - |\mathbf{r} - \mathbf{r}'|/c$. Useful identities: $\partial_r[F] = -\frac{1}{c}[F']$, $\partial_t[F] = [F']$.

4 Dipole Radiation

4.1 The Hertzian dipole

A short oscillating dipole $\mathbf{p} = p_0 e^{-i\omega t} \hat{\mathbf{z}}$ ($d \ll \lambda$) gives $\mathbf{A} = \frac{\mu_0}{4\pi r} [\dot{\mathbf{p}}]$. The fields contain three ranges:

$$\underbrace{\propto 1/r^3}_{\text{near (dipole)}}, \quad \underbrace{\propto 1/r^2}_{\text{induction}}, \quad \underbrace{\propto 1/r}_{\text{radiation}}.$$

Far-field ($r \gg \lambda$) radiation fields:

$$E_\theta = \frac{\sin\theta}{4\pi\epsilon_0} \frac{[\ddot{p}]}{rc^2}, \quad B_\phi = \frac{\mu_0 \sin\theta}{4\pi} \frac{[\ddot{p}]}{rc}, \quad E_\theta = cB_\phi.$$

Radiation is linearly polarized along $\hat{\boldsymbol{\theta}}$ with angular distribution $\propto \sin^2\theta$ (null along the dipole axis). Time-averaged total power:

$$\langle P \rangle = \frac{\mu_0 \omega^4 p_0^2}{12\pi c} = \frac{\omega^4 p_0^2}{12\pi\epsilon_0 c^3}.$$

4.2 Multipole expansion in the far field

Expanding $\mathbf{A}$ in powers of k gives successive terms:

- $m = 0$: electric dipole (ED) radiation (above);
- $m = 1$ antisymmetric: magnetic dipole (MD), $\mathbf{m} = \frac{1}{2} \int \mathbf{r} \times \mathbf{J} dV$;
- $m = 1$ symmetric: electric quadrupole (EQ), tensor $Q_{\alpha\beta} = \int (3x_\alpha x_\beta - r^2 \delta_{\alpha\beta}) \rho dV$.

For comparable size a and current, $\langle P_{MD} \rangle / \langle P_{ED} \rangle \sim (2\pi a / \lambda)^2 \ll 1$, so small magnetic dipoles are weak radiators. The quadrupole power $\propto \omega^6$ (vs. ω^4) and is even weaker for $ka \ll 1$; its pattern is not cylindrically symmetric.

5 Antennas

An antenna's energy loss appears as a **radiation resistance** $R_r = \langle P \rangle / \langle I^2 \rangle$. For a Hertzian dipole,

$$R_r = \frac{2\pi}{3} Z_0 \left(\frac{d}{\lambda} \right)^2 \approx 789 \left(\frac{d}{\lambda} \right)^2 \Omega.$$

Power gain $G(\theta, \phi) = N(\theta, \phi) / [\frac{1}{4\pi} \int N d\Omega]$; for an ED $G_{ED} = \frac{3}{2} \sin^2 \theta$. As a *receiver*, maximum power is delivered to a matched load $R_{\text{load}} = R_r$; the **effective area** obeys the universal, thermodynamically-provable reciprocity relation

$$A_{\text{eff}}(\theta, \phi) = \frac{\lambda^2}{4\pi} G(\theta, \phi),$$

true for *any* antenna (high radiative efficiency $\Leftrightarrow$ high sensitivity). A matched Hertzian dipole re-radiates as much power as it absorbs. **Centre-fed linear antennas**: standing current waves; the half-wave ($L = \lambda/2$) dipole has $R_r \approx 73.1 \Omega$ (easily matched) and a slightly more directional pattern than the ED. **Arrays**: N dipoles give an N -slit-like interference factor; phasing the elements steers the beam (phased-array radar). **Yagi–Uda**: a driven element with reflector and directors gives strong forward directivity.

6 Light Scattering

Radiation incident on a small ($a \ll \lambda$) polarizable particle induces a dipole that re-radiates; cross-section $\sigma = \langle P \rangle / (\text{incident flux})$.

- **Polarization of scattered light**: for unpolarized incidence the scattered degree of polarization at angle α is $P = (1 - \cos^2 \alpha) / (1 + \cos^2 \alpha)$; the dipole-null direction underlies the Brewster condition.
- **Rayleigh scattering** (induced $p_0 = \alpha E_0$): $\sigma \propto \omega^4 \propto \lambda^{-4}$ — the blue sky and sky polarization.
- **Thomson scattering** (free electron): frequency-independent $\sigma_T = \mu_0^2 e^4 / (6\pi m_e^2) = \frac{8\pi}{3} r_e^2 = 6.65 \times 10^{-29} \text{ m}^2$, with classical electron radius r_e . (Compton scattering at high frequency.)
- **Collections of scatterers**: total power $P_{\text{tot}} = P_1(q) |\sum_j e^{i\mathbf{q} \cdot \mathbf{r}_j}|^2$ with scattering vector $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_s$. The **structure factor** $F(\mathbf{q}) = |\tilde{n}(\mathbf{q})|^2$ is the (mod-square) FT of the number density. Uncorrelated (gas): $P_{\text{tot}} = NP_1$. Correlated (crystal): sharp Bragg peaks at reciprocal-lattice vectors. **Critical opalescence** arises from density fluctuations on the scale of λ_{vis} .

7 Special Relativity & Covariant Electrodynamics

Maxwell's equations are not Galilean invariant, motivating SR. Metric $g = \text{diag}(1, -1, -1, -1)$; $\partial^\alpha = (\partial_{ct}, -\nabla)$.

Charge invariance and the 4-current

Electric charge is a Lorentz invariant. Charge conservation plus Lorentz contraction ($\rho = \gamma \rho_0$) shows $J = (c\rho, \mathbf{J})$ is a 4-vector; $\partial_\alpha J^\alpha = 0$ is the (frame-independent) continuity equation.

Covariant Maxwell equations

The 4-potential $A = (\phi/c, \mathbf{A})$ satisfies $\square^2 A = \mu_0 J$. Defining the antisymmetric field-strength tensor $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$,

$$\partial_\alpha F^{\alpha\beta} = \mu_0 J^\beta \quad (\text{ME1, ME4}), \quad \partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0 \quad (\text{ME2, ME3}).$$

The Lorenz gauge $\partial_\alpha A^\alpha = 0$ is itself invariant. The Lorentz force is $dP^\alpha/d\tau = qF^{\alpha\beta}U_\beta$.

Field transformations

$$E'_\parallel = E_\parallel, \quad \mathbf{E}'_\perp = \gamma[\mathbf{E} + \mathbf{v} \times \mathbf{B}]_\perp, \quad B'_\parallel = B_\parallel, \quad \mathbf{B}'_\perp = \gamma[\mathbf{B} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}]_\perp.$$

$\mathbf{E}$ and $\mathbf{B}$ are two aspects of one field. **Invariants:** $|\mathbf{E}|^2 - c^2|\mathbf{B}|^2$ and $\mathbf{E} \cdot \mathbf{B}$.

Magnetism as a relativistic effect

The force between a current-carrying wire and a moving charge can be derived purely from electrostatics + Lorentz contraction, reproducing the magnetic (Lorentz) force — the origin of **spin-orbit coupling** (with the Thomas-precession factor $\frac{1}{2}$).

8 Radiation from Moving Charges

8.1 Uniformly moving charge

Transforming the rest-frame Coulomb field gives an $\mathbf{E}$ directed from the *present* position, flattened into the transverse plane for $\gamma \gg 1$, plus an azimuthal $B_\phi = (v/c^2)E_\perp$. The Poynting flux $\sim 1/r^4$: **a uniformly moving charge does not radiate** (in vacuum).

8.2 Čerenkov radiation

In a medium a charge with $v > c/n$ outruns its own field. A Huygens construction gives a radiating cone at the **Čerenkov angle** $\cos \theta_C = c/(nv) = 1/(\beta n)$. The Frank–Tamm analysis (Fourier solution of the wave equations, modified Bessel functions K_0, K_1) shows energy escapes to infinity only where $\epsilon(\omega) > 1/\beta^2$, giving the characteristic frequency bands (the blue glow).

8.3 Liénard–Wiechert potentials and accelerated charges

For a point charge,

$$\Phi = \frac{1}{4\pi\epsilon_0} \left[\frac{e}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})R} \right]_{\text{ret}}, \quad \mathbf{A} = \frac{\mu_0}{4\pi} \left[\frac{e\mathbf{v}}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})R} \right]_{\text{ret}}.$$

The fields split into **velocity fields** ($\propto R^{-2}$, non-radiative) and **acceleration (radiation) fields** ($\propto R^{-1}$). An accelerating charge radiates (a transverse kink propagates at c).

8.4 Larmor & Liénard formulae

Non-relativistic: **Larmor** $P = \frac{\mu_0 e^2}{6\pi c} |\dot{\mathbf{v}}|^2$. Total radiated power is a Lorentz invariant; using the acceleration 4-vector ($\alpha^2 = \gamma^6 a_\parallel^2 + \gamma^4 a_\perp^2$) gives the relativistic **Liénard formula**

$$P = \frac{\mu_0 e^2}{6\pi c} (\gamma^6 a_\parallel^2 + \gamma^4 a_\perp^2).$$

8.5 Cyclotron and synchrotron radiation

A charge in $\mathbf{B} = B\hat{z}$ moves in a circle at gyrofrequency $\omega_B = eB/(\gamma m)$.

- **Cyclotron** ($v \ll c$, $\gamma \rightarrow 1$): dipole approximation valid; *monochromatic* radiation at ω_B (two quadrature dipoles). Polarization: linear (equatorial), circular (polar), elliptical (between).
- **Synchrotron** ($\gamma \gg 1$): radiation beamed into a forward cone of half-angle $\sim 1/\gamma$ (the $(1 - \beta \cdot \mathbf{n})$ denominator). The observer sees short pulses of duration $\Delta t \sim 1/(\gamma^3 \omega_B)$, giving a broad continuum up to $\nu_s \sim \gamma^3 \omega_B$ (hard X-rays). Important in accelerators (energy loss), X-ray sources and astrophysics (supernovae, radio galaxies).

9 Themes from the Examples Sheet

The examples are examinable; several introduce methods worth knowing (detailed in the companion derivations document):

- **Jones-matrix problems** (Q3, Q6–Q8): deriving J_θ , crossed polarizers with an inserted $\lambda/4$ plate, intensity vs. angle.
- **Waveplate design** (Q4–Q5): minimum thickness $d_{\min} = \lambda/(4|n_e - n_o|)$.
- **Cavity modes** (Q11): standing-wave frequencies for uniaxial, optically-active and Faraday media via polarization eigenstates.
- **Plasma in a field / helicon modes** (Q12): low-frequency dispersion $\omega = c^2 k^2 \omega_c / \omega_p^2$ and group velocity.
- **Coherence applications** (Q13–Q15): expanding-shell velocity from visibility, Cd-red coherence length, the lateral-coherence sinc for a slit source $\gamma(d) = \text{sinc}(\pi w d / f \lambda)$.
- **Vector potential of a magnetic dipole** (Q16–Q17): reconstructing $\mathbf{A}$ and $\mathbf{J}$ from given $\mathbf{B}$; testing whether a field is physical ($\nabla \cdot \mathbf{B} = 0$).
- **Rotating dipoles / pulsars** (Q19–Q20): quadrature decomposition; spin-down law $\omega \dot{\omega} = 3\dot{\omega}^2$ and the Crab pulsar field.
- **Loop antenna** (Q21): $R_r = \mu_0 \omega^4 a^4 / (6\pi c^3)$, $G = \frac{3}{2} \sin^2 \theta$, combined scattering+absorption cross-section.
- **Atmospheric / sky scattering** (Q22–Q23): Rayleigh estimates, $\sim 45\%$ sky polarization at noon.
- **Relativistic problems** (Q24–Q31): moving dielectric slab and refractive index $n_v = (n + \beta)/(n\beta + 1)$; covariant equation of motion; transforming capacitor and conductor fields; force between co-moving charges; EM invariants and transformed radiation energy density; 1 GeV synchrotron estimates.

End of summary. See “Electrodynamics & Optics — Key Derivations” for the worked mathematics behind the boxed results above.